Stage 2 — Questions and Answers

Overall Survival Prediction in Adult Myeloid Leukemia

Q1. One Kaplan–Meier step by hand

At t = 0.01644 years, row 3 has n = 3,107 patients at risk and d = 1 death. Using row 2's survival probability S = 0.99904:

$$S(0.01644) = 0.99904 \times (1 - 1/3107)$$

$$= 0.99904 \times (1 - 0.00032185)$$

$$= 0.99904 \times 0.99967815$$

$$= \mathbf{0.99872} \text{ (rounded to five decimal places).}$$

The at-risk count fell from 3,114 to 3,107, a loss of 7 patients. Of those, 2 died at row 2's time and the other 5 were censored before row 3. Deaths reduce the Kaplan–Meier survival probability, whereas censoring removes patients from the risk set without directly lowering the curve. The first row starts at 3,120 instead of 3,173 because 53 patients were censored at time zero.

Q2. How much do late deaths count?

Initially 7 of the 9 comparable pairs were concordant. Correcting the (E, F) ranking increases the number to 8, while the total number of comparable pairs stays at 9.

$$\text{Harrell's } C = 8/9 = \mathbf{0.8889} \text{ (previously } 7/9 = 0.7778).$$

The IPCW numerator gains E's weight, 3.5156, and the denominator stays 13.2031:

$$\text{IPCW } C = (8.1250 + 3.5156)/13.2031$$

$$= 11.6406/13.2031 = \mathbf{0.8817}.$$

Harrell's C improves by 1/9 = 0.1111, while IPCW C improves by 3.5156/13.2031 = 0.2663. E's weight is high because $G(5) \approx 0.5333$, giving $1/G(5)^2 \approx 3.5156$; by comparison A's weight is 1 and C's is 1.5625. Thus, the challenge metric penalizes mistakes involving later observed deaths more heavily.

Q3. Why does TP53 rank so poorly on its own?

Of the 3,173 patients, 368 are TP53-mutated and 2,805 are not. The mutation share is:

$$p = 368/3173 \approx \mathbf{0.11598} \text{ (11.60\%).}$$

For two randomly selected patients, the probability that their binary TP53 scores match is:

$$P(\text{same}) = p^2 + (1 - p)^2$$

$$= (0.11598)^2 + (0.88402)^2$$

$$= \mathbf{0.79494} \approx \mathbf{79.49\%}.$$

About 79.5% of random patient pairs have the same TP53 value and therefore cannot be ranked by this variable alone; tied risk scores contribute only one-half to the C-index. Even with perfect ordering of all pairs with different TP53 status, the approximate ceiling for random pairs would be $0.5 \times 0.79494 + 1 \times 0.20506 \approx 0.60253$. The actual IPCW C-index for TP53 is 0.5535, despite its strong log-rank test (chi-square = 244.6).

The log-rank test asks whether survival differs between groups; the C-index asks whether individual patients can be ordered correctly. BM_BLAST is continuous and has fewer ties, giving a higher IPCW C-index of 0.6157. Stage 3 should therefore combine clinical, cytogenetic and mutation features into a multivariable risk score rather than rely on TP53 alone.